

Praneth Mutyala

Generative AI Engineer | LLM Engineer

+1 8624057154 | pranethm65@gmail.com | [Linkedin](#)

PROFESSIONAL SUMMARY

Senior AI Engineer with 5 years of experience delivering agentic AI solutions, document automation, and large-scale workflow orchestration for finance and healthcare sectors. Expert in building production-ready AI applications, including agentic frameworks, RAG systems, and intelligent document processing using cloud-native technologies.

EXPERIENCE

Northern Trust

Sep 2025 - Present

Generative AI Engineer

Remote

- Designed and deployed production AI services using Python, FastAPI, Docker, Cloud Run, and Vertex AI for scalable fraud detection and credit risk workflows.
- Developed semi-autonomous AI agents for intelligent document analysis and workflow automation within fraud and compliance operations.
- Developed scalable feature engineering and data processing pipelines using PySpark, BigQuery, and Dataproc to support machine learning models on large banking datasets.
- Implemented real-time inference pipelines using Pub/Sub, Dataflow, and Vertex AI Endpoints for low-latency AI predictions, supporting agentic task orchestration and workflow integration.
- Built graph-based fraud analysis using Neo4j to identify suspicious relationships between accounts, merchants, and devices.
- Collaborated with data engineers, fraud analysts, and business stakeholders to deliver secure, production-ready AI solutions aligned with regulatory requirements.

Humana

May 2024 - Aug 2025

AI/ML Engineer

Remote

- Developed Hybrid RAG applications using Azure OpenAI, LlamaIndex, and FAISS to provide clinicians with contextual access to medical guidelines and historical patient information.
- Designed agentic AI workflows for document analysis, clinical decision support, and data extraction using LangChain and LangGraph.
- Integrated Model Context Protocol (MCP) to enable communication between AI agents, enterprise services, and retrieval systems and orchestrated multi-step automation pipelines.
- Built NLP pipelines using BioBERT, Hugging Face Transformers, and spaCy to extract clinical entities from physician notes and medical documents.
- Deployed scalable AI applications using FastAPI, Docker, Azure Kubernetes Service (AKS), and MLflow, supporting production deployment and monitoring.
- Implemented observability and governance using Prometheus, Grafana, LangSmith, and HIPAA-compliant security practices to monitor AI system performance.

Birlasoft

Apr 2021 - Jul 2023

Python Developer

Hyderabad

- Developed Python and SQL data processing pipelines for customer analytics and large-scale banking datasets.
- Built machine learning models for customer churn prediction and segmentation using Scikit-learn and feature engineering techniques.
- Processed enterprise datasets using PySpark, Hadoop, and Hive to support scalable analytics and reporting.
- Created business dashboards and visualizations using Tableau and Matplotlib to communicate analytical insights.
- Collaborated with cross-functional Agile teams to deliver data-driven solutions supporting customer engagement and business decision-making.

TECHNICAL SKILLS

- Visualization & Analytics:** Tableau, Power BI, Matplotlib
- Cloud Platforms:** Google Cloud Platform (Vertex AI, BigQuery), Microsoft Azure (Azure OpenAI, AKS), AWS (EC2, S3, Lambda, Step Functions)
- MLOps & LLMops:** Docker, MLflow, Git, GitHub, Jenkins, CI/CD
- Agentic AI & Automation:** Agentic Frameworks, LangChain, LangGraph, Model Context Protocol (MCP), AI Workflow Tools, Reverse Engineering Workflows
- Programming Languages:** Python, SQL, Java, R, TypeScript
- Generative AI & LLMs:** Azure OpenAI, Vertex AI, LlamaIndex, Hugging Face Transformers, Retrieval-Augmented Generation (RAG), Embedding Models
- Big Data & Data Engineering:** PySpark, Apache Spark, Databricks, Apache Airflow, BigQuery, Snowflake
- Observability & Monitoring:** LangSmith, Prometheus, Grafana
- Deep Learning:** PyTorch, TensorFlow, BERT, LSTM, Transfer Learning, XGBoost, LightGBM, Feature Engineering
- Databases & Vector Stores:** PostgreSQL, MySQL, MongoDB, FAISS, Pinecone
- Backend Development:** FastAPI, REST APIs, Load Balancing, System Design

EDUCATION

New Jersey Institute of Technology | Master of Science, Data Science

Newark, New Jersey

Vellore Institute of Technology | Bachelor of Science, Computer Science

Amaravathi, India

CERTIFICATIONS

- AWS Certified Developer – Associate
- Microsoft Azure Fundamentals (AZ-900)